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NORMAL VARIABILITY OF THE BRAIN-STEM AUDITORY EVOKED RESPONSE IN YOUNG AND OLD ADULT SUBJECTS*

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The brain-stem auditory evoked response is the seven peaked waveform recorded from the scalp in the first 10 msec following presentation of a click stimulus to the ear. The generators of the vertex positive peaks have been related to sequential components of the auditory pathway. Laboratory and clinical evidence now suggest the following wave origins: (1) auditory nerve, (2) cochlear nucleus, (3) superior olivary complex, (4) midbrain, possibly nucleus of the lateral lemniscus, (5) inferior colliculus, (6) medial geniculate and (7) possibly auditory radiation from the thalamus to temporal cortex. (Buchwald and Huang 1975; Jewett 1970; Jewett et al. 1970; Jewett and Williston 1971; Lev and Sohmer 1972; Picton et al. 1974; Sohmer and Feinmesser 1967; Sohmer et al. 1974; Starr and Hamilton 1976; Stockard and Rossiter 1977).

The amplitude and latency of the response components vary with stimulus intensity (Hecox and Galambos 1974; Jewett et al. 1970; Lev and Sohmer 1972; Lieberman et al. 1973b). This variation has been used clinically to determine accurately peripheral auditory acuity (Jewett and Williston 1971; Lieberman et al. 1973a; Schulman-Galambos and Galambos 1975; Sohmer and Feinmesser 1973). The response interpeak conduction times vary with structural and physiological disorders of the brain-stem which affect the auditory pathway (Jones et al. 1976; Lev and Sohmer 1972; Sohmer et al. 1974; Starr and Hamilton 1976; Stockard et al. 1976a; Stockard and Rossiter 1977). This disruption of waveform has been used clinically to demonstrate objectively brain-stem dysfunction (Robinson and Rudge 1975; Selters and Brackmann 1977; Starr 1976; Starr and Achor 1975; Stockard et al. 1976b; Thornton 1975a).

Unfortunately, the published studies with normal values for peak latencies and for interpeak conduction times are often not comparable because of variations in stimulus technique (Hecox and Galambos 1974; Lieberman et al. 1973b; Salamy et al. 1975; Selters and Brackman 1977; Starr and Achor 1975; Stockard and Rossiter 1977; Terkildsen et al. 1973; Thornton 1975b, 1975c). Also, there have been no detailed studies published on interear symmetry of response to monaural testing or on the effects of aging or of decreased intensity of stimulation upon the interpeak conduction times.

This study determined normative peak latency and interpeak conduction time values for monaural stimulation in young and old adult subjects at three different stimulus intensity and rate combinations. Comparisons are made of the quality and reproducibility of the responses and of the variations that occur with changes in stimulus technique.

Method

Twenty-five young (mean age 25.1 years, range 17—33 years) and 25 old (mean age 61.7 years, range 51—74 years) subjects with no history of hearing deficit or neurologic

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disease were tested. Hearing thresholds to click stimuli were within 6 dB of the laboratory standard for the young group and within 14 dB for the old group.

Responses were recorded by the method of Hecox and Galambos (1974). A Grass S9 stimulator delivered a 0.1 msec square wave through a Hewlett-Packard 3550 Decibel Attenuator to one side of TDH-39 or Mura SP-205 earphones*. Silver disk electrodes were placed in conduction paste on both mastoids and the vertex. A vertex to mastoid derivation ipsilateral to the stimulated ear with contralateral mastoid ground was amplified by two Grass P-15 amplifiers in tandem with a total gain of 100 000 and a bandwidth (at 3 dB down) of 100—3000 Hz. Occasionally, if muscle activity was present, a narrower bandwidth of 300—3000 Hz was used. The responses were summated in a Nicolet 1072 averager and written out with a Hewlett-Packard 7035B X-Y plotter.

Testing was performed at the ambient background noise of the laboratory. The laboratory click threshold was set at the average threshold of 5 young adults with audiometrically demonstrated normal hearing. Click stimuli were delivered at 60 dB sound level above this threshold at stimulation rates of 10/sec and 30/sec to compare the effects of change in rate of stimulation. A third trial at 30 dB sound level and 30/sec rate was given to compare the effects of decreased intensity of stimulation.

Four responses at each stimulating condition were obtained using an adequate number of stimuli to obtain clearly reproducible waveforms. This required from 1024 to 8192 stimuli per response. The four replications were then summated and written out. Wave peaks were identified if present in 3 of the 4 replications at approximately the same latency. The peak latencies were determined to the nearest 0.1 msec from the X-Y plot of the final summated waveform. Amplitude of wave 5 was measured from the positive peak of wave 5 to the following negative valley. Peak latencies and interpeak conduction times were analysed between groups using one way analysis of variance.

Measuring peak latencies from the X-Y plot introduced a fixed error in all values. The plot was set to a 10 msec grid, while the total time encompassed in the Nicolet average is 10.24 msec. For this paper all numerical data were therefore multiplied by 1.024 to obtain their true values.

Results

There were several difficulties encountered in recording and analysing these complex waveforms. Muscle activity in tense patients could distort or completely obliterate the response, producing a variability of wave amplitude on repeat testing (Fig. 1). In relaxed, cooperative subjects there were often quite marked differences in the waveforms elicited from the two ears (Fig. 2). Prominent asymmetries were seen in 10 subjects. Less marked interear differences were seen in an additional 10. The wave shape was not affected by change in the rate of stimulation, but it was altered drastically by change in stimulus intensity (Figs. 1 and 2). The decrease in stimulus intensity from 60 dB to 30 dB produced a lower amplitude response with a lower signal to noise ratio. Many more stimuli were needed at the lower intensity to produce replicable responses (Fig. 1).

Wave 1 was often low in amplitude and difficult to differentiate from background noise without replications (Fig. 3). A reproducible wave 1 could not be obtained in 3% of the 60 dB trials or in 23% of the 30 dB trials. There were no differences between 10/sec and 30/sec rates of stimulation in the amplitude or distinctiveness of wave 1.

Waves 2 and 4 were frequently absent from one or both sides. In both young and old subjects wave 2 was absent in 20% and wave 4 in 33% of trials at 60 dB, regardless of

* The earphones produced equivalent responses in 5 subjects tested with both types.

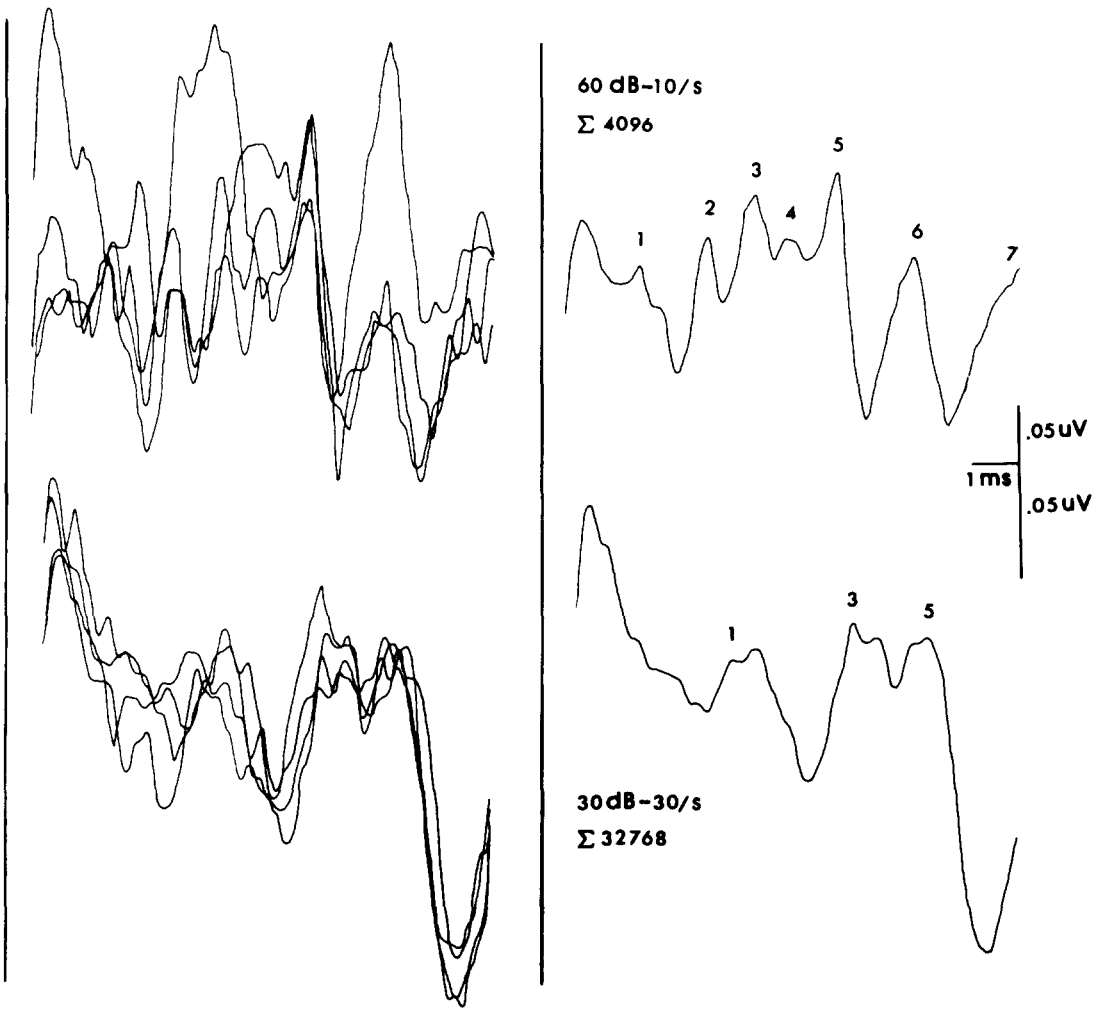
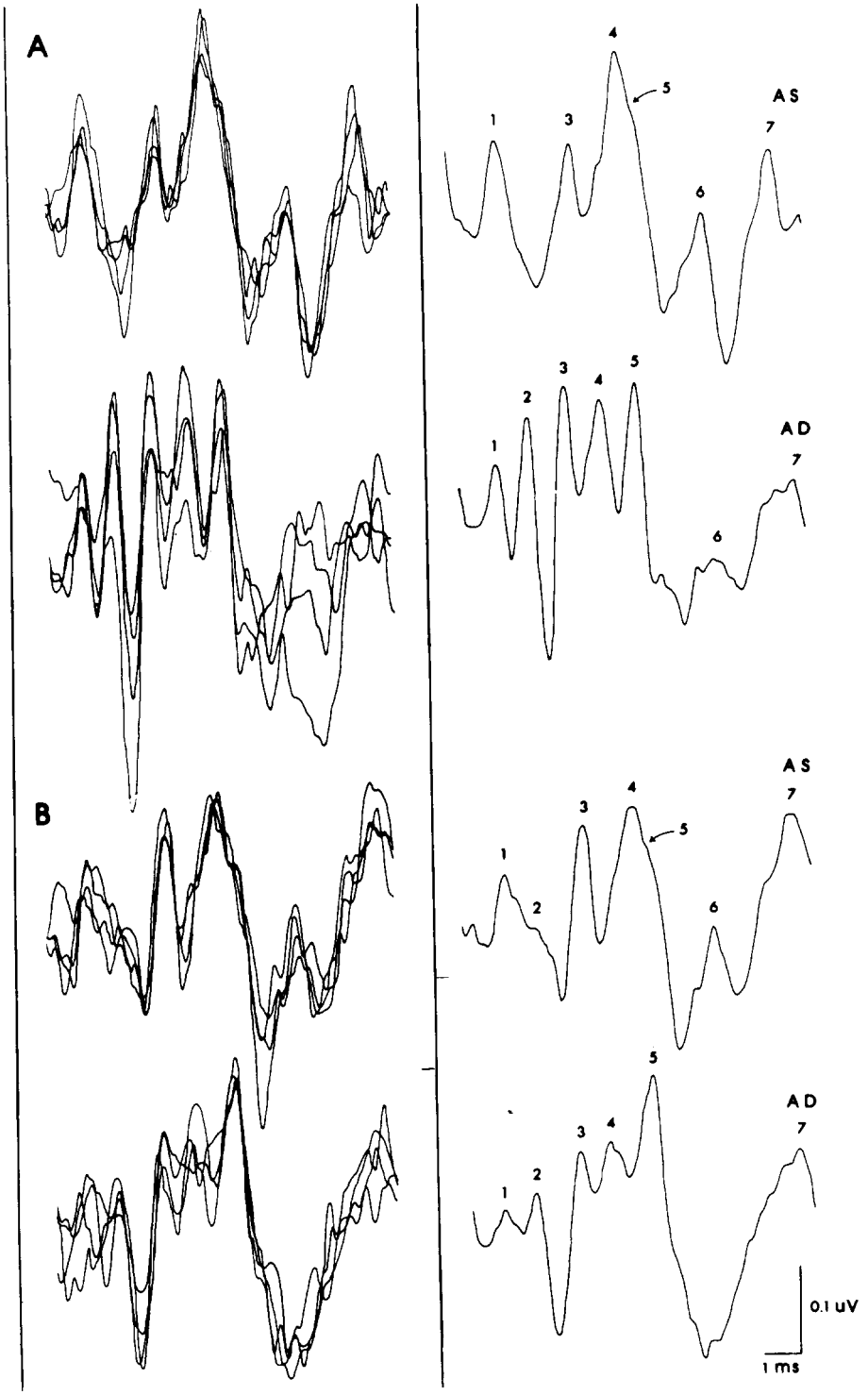


Fig. 1. The 60 dB-10/sec responses (top) show the variation in amplitude and waveform which can occur between replications with the presence of muscle activity. The 7 vertex positive peaks can be identified in all replications. The peak latencies are constant while the amplitude relationships are not. The 30 dB-30/sec responses (bottom) show the change in waveform which occurs with decreased intensity of stimulation. Note that the calibration is different and that 8 times the number of stimuli were needed to produce replicable waveforms. This and all following responses have been traced for better reproduction. Stimulus onset is at the vertical line and the stimulus artifact has been omitted. Vertex positive is upward.

stimulating frequency. At 30 dB wave 2 was absent in 62% and wave 4 in 60% of trials.

Waves 3 and 5 were the most constant and reproducible peaks being seen in all subjects at 60 dB stimulation. At 30 dB stimulation wave 3 was absent in 16% and wave 5 in 3% of trials. With all types of stimulation wave 5 appeared in a fused form with wave 4 in 10% of trials, often as a break on the downslope of wave 4 (Fig. 4). When the identity of wave 5 was in doubt, stimulation at a slightly lower sound level would separate wave 4 and 5 adequately to determine wave 5 peak latency.

Waves 6 and 7 often varied widely in


A
B
AS
AD
AS
AD
1
2
3
4
5
6
7
0.1 uV
1 ms

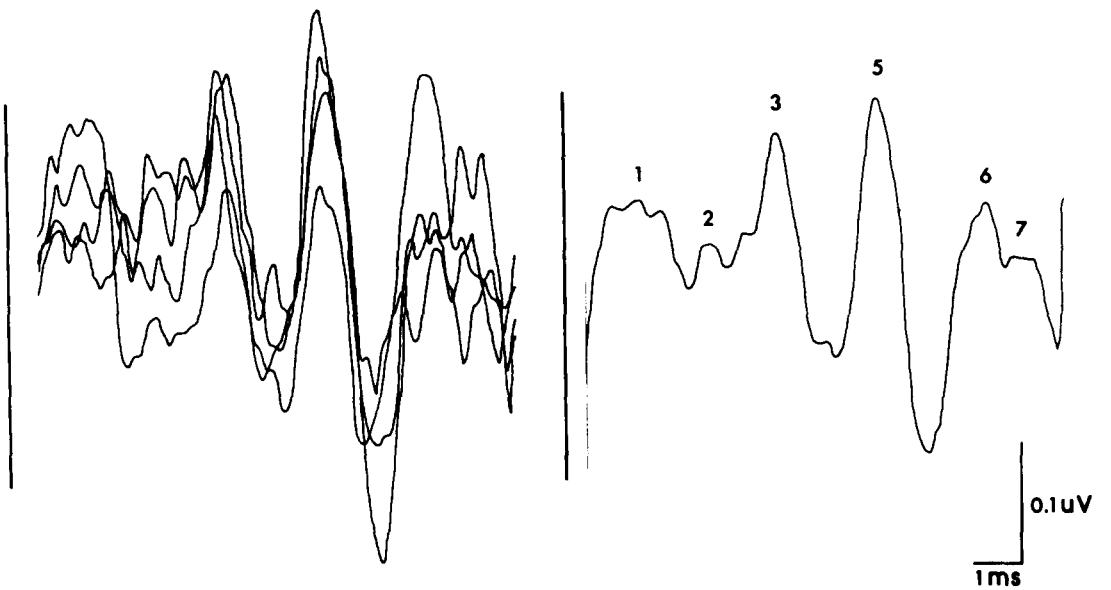


Fig. 3. Wave 1 is frequently low in amplitude and difficult to distinguish from background noise. In this response wave 1 is chosen as the peak which shows the best reproducibility. The peak occurring just prior to wave 1, between 1.0 and 1.5 msec, was often seen in different subjects. When the intensity of stimulation is decreased this peak remains fixed in position while wave 1 increases in latency. It is uncertain whether this represents a component of the stimulus artefact or a late cochlear potential.

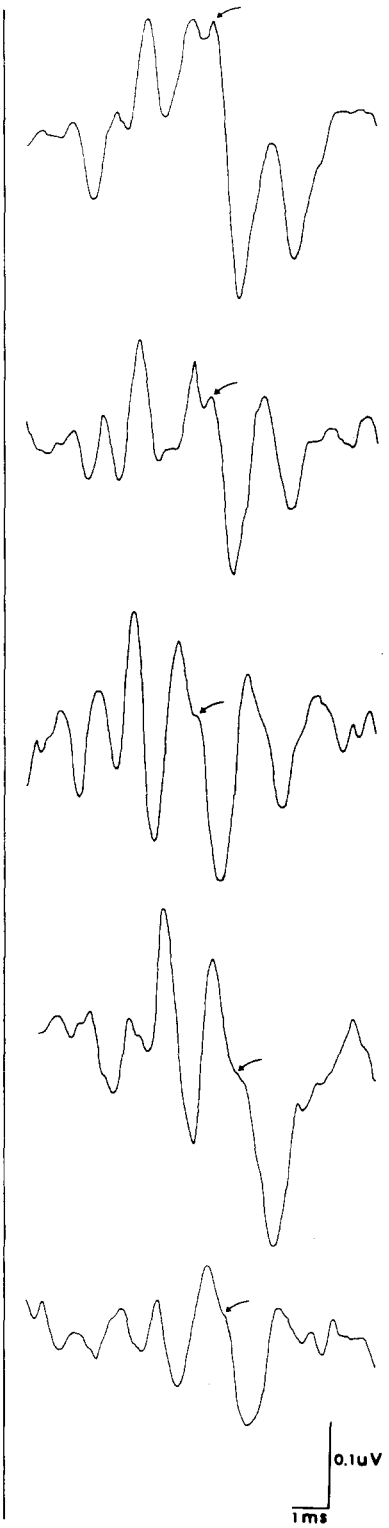
waveform from one replication to another (Fig. 2). These waves were slightly better demonstrated by 10/sec than by 30/sec stimulation.

Mean peak latencies for both ears combined are presented in Table I. There were no statistical differences in the latencies for left and right ear responses in any of the trial categories. The 99% confidence intervals for the latencies of waves 1—5 were approximately 1 msec in width while those for waves 6 and 7 were over 2 msec. In individual subjects, interear differences in latency were within 0.4 msec for waves 1 through 5 in 95%. Waves 6 and 7 showed a greater interear difference of 1.0 msec in 95% of subjects.

The mean peak latencies at 60 dB-10/sec and 60 dB-30/sec stimulation were statistically different at waves 1 and 4 ($P < 0.05$) and at waves 3, 5 and 6 ($P < 0.001$). All 30 dB-30/sec latencies were significantly increased compared to the 60 dB-30/sec values ($P < 0.001$). This difference resided entirely in the increased latency of wave 1. When all responses were adjusted to an equal wave 1 the subsequent latencies were seen to be similar, except for the 30 dB-30/sec wave 2 responses. These were relatively decreased in latency compared to the 60 dB-30/sec trials in the old subjects ($P < 0.01$) but only suggestively changed in the young subjects ($P < 0.10$).

Interwave conduction times were calculated

Fig. 2. Responses obtained from the left (AS) and right (AD) ears of a normal young subject. (A) 60 dB-10/sec stimulation, (B) 60 dB-30/sec stimulation. Note that change in frequency of stimulation does not significantly alter the waveforms. The responses show marked differences in waveform between the two ears. Wave 2 is absent in A-AS. Wave 5 is present in a fused form with wave 4 in A-AS and B-AS (see Fig. 4). Wave 6 is poorly replicable in A-AD and absent in B-AD. Asymmetries in response such as this make comparisons of waveform clinically unrewarding.



for a variety of wave combinations (Table II). The mean conduction times for left and right ear stimulation were not significantly different at any of the stimulus combinations. 95% of individual subjects showed conduction times to be within 0.5 msec between the two ears for all stimulus combinations.

When conduction times were compared (Fig. 5) the time from wave 3 to 5 was found to be equal in all test conditions in both young and old subjects. That is, no statistical differences were noted between their distributions. The time from wave 1 to 3, and therefore from wave 1 to 5, increased slightly with increased frequency of stimulation and increased age. Wave 2 appeared to shift towards wave 1 with decreased intensity of stimulation, with a concomitant increase in wave 2—3 and 2—5 times. Wave 5 to 6 time appeared to increase slightly with a decrease in intensity. Wave 5 to 7 time was unaffected by change in stimulus condition or age.

The variance of the conduction times generally increased with a decrease in the intensity of stimulation. This reflects the lower signal to noise ratio and the subsequent increased variability of the response found with low intensity stimulation.

One tailed 95% and 99% confidence intervals were prepared for conduction times for 60 dB stimulation at both 10/sec and 30/sec, with suitable correction factors to be added when used in older subjects (Table III). No correction factor is necessary for change in stimulus intensity.

Wave 5 amplitude was evaluated to check the impression that it is a variable and unreliable measure of the response waveform (Table IV). In all stimulus combinations the 99% confidence interval encompassed zero volts. This indicates that a low amplitude

Fig. 4. Wave 5 may occur in a fused form with wave 4 as shown in responses from 5 different subjects. It may take a large number of stimuli to show the break in the downslope of wave 4, which is wave 5, reproducibly. When in doubt, stimulation at a lower intensity will usually separate the fused wave into its two components.

TABLE I

Wave Peak Latencies for all stimulus conditions, right and left ears combined. Times given in milliseconds. S.D. = standard deviation.

Waves	60 dB-10/sec		60 dB-30/sec		30 dB-30/sec	
	Mean (S.D.)	Number	Mean (S.D.)	Number	Mean (S.D.)	Number
Young subjects						
1	1.87 (0.18)	47	1.96 (0.21)	47	3.16 (0.36)	40
2	2.88 (0.20)	38	2.98 (0.28)	37	4.01 (0.36)	21
3	3.83 (0.20)	50	4.01 (0.18)	50	5.18 (0.31)	46
4	5.06 (0.23)	43	5.19 (0.28)	38	6.36 (0.36)	18
5	5.82 (0.25)	50	6.01 (0.25)	50	7.13 (0.33)	49
6	7.37 (0.46)	50	7.70 (0.45)	46	8.95 (0.40)	36
7	9.05 (0.47)	48	9.24 (0.42)	38	—	—
Old subjects						
1	2.17 (0.27)	50	2.16 (0.31)	50	4.21 (0.61)	37
2	3.36 (0.31)	41	3.28 (0.33)	43	4.96 (0.62)	17
3	4.35 (0.26)	50	4.45 (0.35)	50	6.28 (0.51)	38
4	5.43 (0.34)	24	5.54 (0.35)	30	7.35 (0.41)	22
5	6.16 (0.26)	49	6.37 (0.27)	50	8.21 (0.47)	48
6	7.95 (0.38)	50	8.18 (0.42)	48	9.50 (0.44)	13
7	9.30 (0.51)	41	9.38 (0.48)	33	—	—

TABLE II

Interpeak conduction times for all stimulus conditions, right and left ears combined. Times given in milliseconds. S.D. = standard deviation.

Waves	60 dB—10/sec		60 dB—30/sec		30 dB—30/sec	
	Mean (S.D.)	Number	Mean (S.D.)	Number	Mean (S.D.)	Number
Young subjects						
1—3	1.97 (0.16)	47	2.08 (0.21)	47	1.94 (0.33)	40
3—5	1.97 (0.20)	50	2.00 (0.21)	50	1.93 (0.26)	46
1—5	3.94 (0.22)	47	4.05 (0.24)	47	3.94 (0.39)	41
1—2	1.03 (0.21)	36	1.09 (0.25)	36	0.92 (0.37)	17
2—3	1.03 (0.32)	38	1.02 (0.30)	37	1.21 (0.48)	21
2—5	2.90 (0.24)	38	2.95 (0.38)	37	3.06 (0.49)	21
5—6	1.58 (0.39)	50	1.70 (0.38)	46	1.87 (0.28)	36
5—7	3.27 (0.42)	48	3.27 (0.42)	39	—	—
Old subjects						
1—3	2.19 (0.27)	50	2.30 (0.34)	50	2.18 (0.43)	29
3—5	1.82 (0.21)	50	1.91 (0.24)	50	1.89 (0.28)	36
1—5	4.00 (0.32)	50	4.21 (0.34)	50	4.03 (0.42)	37
1—2	1.22 (0.23)	41	1.14 (0.25)	42	0.92 (0.30)	15
2—3	0.99 (0.21)	42	1.19 (0.32)	43	1.48 (0.47)	16
2—5	2.82 (0.32)	42	3.11 (0.36)	43	3.26 (0.48)	17
5—6	1.78 (0.34)	50	1.82 (0.32)	48	1.76 (0.30)	12
5—7	3.15 (0.53)	42	3.11 (0.44)	33	—	—

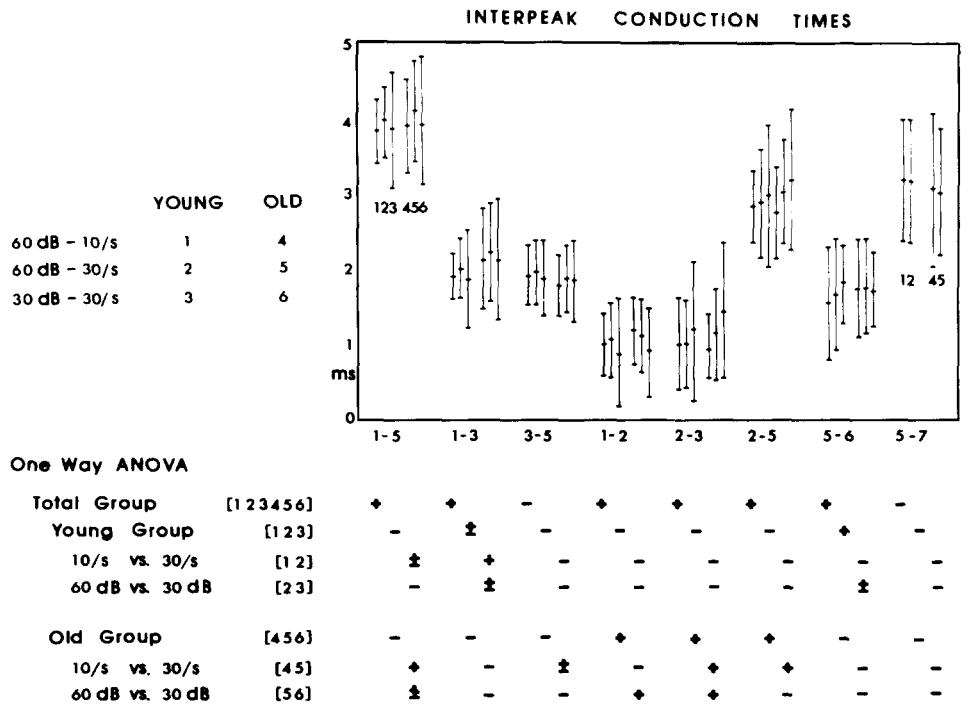


Fig. 5. Mean interpeak conduction times ± 2 standard deviations are plotted for each subject/stimulus condition. One way analysis of variance was performed for each group shown to test whether the components of the group were drawn from the same or different populations. Differences were accepted (+) if $P < 0.01$, considered suggestive (±) if P 0.05 < 0.01 and rejected (-) if $P > 0.05$.

TABLE III

Interpeak conduction time normal range for young subjects. For old subjects add the times noted at the bottom of the table. The times are in milliseconds and represent the upper limits of one tailed confidence intervals. Times above these limits are longer than 95 and 99% of the normal population respectively. Interear differences should be within 0.5 msec.

Waves	60 dB-10/sec		60 dB-30/sec	
	95%	99%	95%	99%
1—3*	2.23	2.34	2.43	2.58
3—5	2.30	2.44	2.36	2.51
1—5*	4.30	4.46	4.46	4.64
1—2**	1.38	1.55	1.52	1.70
2—3	1.58	1.81	1.54	1.76
2—5**	3.31	3.49	3.58	3.87
5—6	2.22	2.51	2.32	2.60
5—7	3.98	4.29	3.97	4.29

* Add 0.3 msec. ** Add 0.2 msec.

TABLE IV

Wave 5 amplitude in all stimulus conditions. The lower limit of the 2 tailed confidence interval for each condition is given to show that the 99% interval encompasses zero volts. Amplitudes are in microvolts. S.D. = standard deviation.

Stimulus conditions	Mean (S.D.)	Number	Confidence interval	
			95%	99%
Young subjects				
60 dB-10/sec	0.29 (0.12)	45	0.05	-0.02
60 dB-30/sec	0.28 (0.11)	43	0.06	-0.01
30 dB-30/sec	0.23 (0.10)	41	0.03	-0.03
Old subjects				
60 dB-10/sec	0.32 (0.12)	49	0.07	-0.01
60 dB-30/sec	0.27 (0.12)	49	0.03	-0.05
30 dB-30/sec	0.11 (0.07)	48	0.03	-0.07

wave 5 cannot be stated to be abnormal purely on the basis of its size. However, in no subject was wave 5 smaller than wave 1.

Discussion

The brainstem auditory evoked response shows great promise for the objective demonstration of neurologic abnormality. However, accurate interpretation of changes in the response waveform must be based on a knowledge of its intrinsic variability.

Technical variables must be minimized. The amplifying method used in this study is not recommended. The Grass P-15 amplifiers, when run in tandem, are prone to develop unstable signals and artifactual ringing of the filters. The Grass Company now advises against their use in this fashion (pers. comm.). Newer systems with better electronic and human engineering are rapidly becoming available for general use.

Response waveforms and amplitudes change drastically in the presence of muscle artifact and with decreased stimulus intensity. The amplitude of wave 5 is so variable that the 99% confidence interval encompasses zero volts. Therefore, individual peak amplitudes are not reliable measures of normality. However, the relative amplitude of waves may be clinically useful. This study confirms the observation of Starr and Achor (1975) that wave 5 is normally larger than wave 1.

The response waveform may also be markedly asymmetrical to left and right ear stimulation. Waves 2, 4, 6 and 7 may be absent unilaterally or bilaterally. Waves 6 and 7 may also show marked individual differences in latency between the ears compared to waves 1—5.

Peak latency values are independent of changes in waveform or amplitude. However, these values are affected by changes in the intensity of stimulation (Hecox and Galambos 1974; Jewett et al. 1970; Lev and Sohmer 1972; Lieberman et al. 1973b). The actual latency values found in any laboratory depend upon the subjectively determined laboratory click threshold and upon the background noise present during the recording. Because of these variables laboratories may differ in their peak latency normative values.*

The interpeak conduction times should not vary between laboratories using the same stimulus rate because they are not dependent

TABLE V

Comparisons of conduction times reported from 4 laboratories using the same technique of 60 dB-10/sec stimulation. Lab. 1, Starr and Achor (1975); Lab. 2, Stockard and Rossiter (1977); Lab. 3, Chiappa, K. (pers. comm.); Lab. 4, this paper. A: Conduction times as computed from reported mean peak latencies. B: Conduction times as reported directly, Mean (S.D.) Times are in milliseconds, S.D. = standard deviation.

A.

Waves	Lab 1	Lab 2	Lab 3	Lab 4	
				Young	Old
1—3	2.3	2.2	2.1	2.0	2.2
3—5	1.7	1.8	1.9	2.0	1.8
1—5	4.0	4.0	4.0	4.0	4.0
1—2	1.2	1.1	1.0	1.0	1.2
2—3	1.1	1.1	1.1	0.9	1.0
2—5	2.8	2.9	3.0	2.9	2.8
5—6	1.5	1.7	1.6	1.6	1.8
5—7	3.3	3.3	—	3.3	3.1

B.

Waves	Lab 2	Lab 3	Lab 4	
			Young	Old
1—3	2.1 (0.3)	2.1 (0.1)	2.0 (0.2)	2.2 (0.3)
3—5	1.9 (0.2)	1.9 (0.2)	2.0 (0.2)	1.8 (0.2)
1—5	4.0 (0.2)	4.0 (0.2)	3.9 (0.2)	4.0 (0.3)
5—6	1.6 (0.2)	1.5 (0.3)	1.6 (0.4)	1.8 (0.3)

* Schulman-Galambos, C. and Galambos, R. Recording the Brainstem Evoked Response: A Manual 1975 unpublished.

upon stimulus intensity. The conduction times found in this study are remarkably similar to those from other laboratories using the same technique (Table V).

Because interpeak conduction times are independent of stimulus intensity, these normal values can be applied to patients with conductive hearing loss. This greatly simplifies the application of brainstem response testing in clinical neurology*.

As waves 1, 3 and 5 are the most reliable and reproducible peaks, the conduction times from waves 1 to 5, 1 to 3 and 3 to 5 appear to be the most clinically useful values. Wave 2 seems to vary slightly with a decrease in stimulus intensity. Conduction times taken from wave 2 would therefore change to an unknown degree in patients with hearing loss. The variability of waves 6 and 7 is such that conduction times involving these waves must be interpreted with great caution. Wave 6 may be absent and wave 7 may be mistakenly identified as wave 6. This would produce an artifactually prolonged wave 5 to 6 conduction time. The 99% confidence limit for wave 5 to 7 conduction time places wave 7 at or beyond the 10 msec timebase of the usual recording situation.

Stimulus rate does affect the interpeak conduction time. In general, 10/sec stimulation gives better waveform reproducibility and somewhat smaller variance of conduction times than does 30/sec. However, in the presence of a hearing deficit or of muscle artifact, a much larger number of stimuli may be necessary to produce replicable responses. In this case, 30/sec stimulation is useful to decrease the duration of the recording session. 30/sec stimulation does result in longer wave 1 to 3 and 1 to 5 conduction times. These times are also increased in older subjects. Appropriate corrections in the normative tables must be made for these variables.

One tailed confidence limits are presented for the conduction times because of the assumption that only an increase in conduction time can definitely be related to a physiologic dysfunction such as desynchronization of nerve volleys or slowing of nerve impulse velocity. An asymmetry of conduction times of greater than 0.5 msec may indicate an abnormality on the slower side even if the conduction times lie within the normal ranges.

At the present (see introductory paragraph), a delay in the wave 1—3 portion of the response would seem to indicate an abnormality in the pontine-medullary region, while a delay in the wave 3—5 portion would suggest dysfunction in the midbrain-pontine region. Delay in the wave 1—5 time without specific prolongation of either of its components should be considered abnormal, but without specific localizing value as to the level of the brain-stem involved.

Summary

The brain-stem auditory evoked response was studied under three combinations of intensity and rate of click stimulation in 25 young and 25 old adult subjects. The responses show a great intra and inter subject variability. Waves 1, 3 and 5 are constant and reproducible markers of the response while waves 2, 4, 6 and 7 are variable and frequently asymmetrical or absent. Wave peak latencies increase with an increase in stimulation rate, a decrease in stimulus intensity and an increase in age. Interpeak conduction times, except those involving wave 2, are unaffected by a change in stimulus intensity. Wave 1—3 time increases with an increase in stimulus rate and an increase in age while wave 3—5 time is not affected by any change in stimulus parameters or age. Wave amplitude alone is not a reliable measure of normality.

Normal values for peak latencies, inter-

* A number of patients with conductive and/or sensorineural hearing loss have shown normal interpeak conduction times in this laboratory. However, no specific study of this problem has been made, nor has any been published to my knowledge to date.

peak conduction times and interear symmetry are presented. One tailed 95 and 99% confidence intervals for normal interpeak conduction times are presented with appropriate corrections for stimulus rate used and patient age. No corrections need to be made for changes in stimulus intensity. This suggests that no correction is necessary when studying patients with conductive type hearing deficits. Laboratories using the same stimulation technique appear to produce equivalent normative conduction time values.

Résumé

Variabilité normale de la reponse evoqué auditive du tronc cerebral chez des sujets adultes jeunes et agés

La réponse évoquée auditive du tronc cérébral a été étudié dans trois conditions d'intensité et de vitesse de stimulation par clicks sur 25 sujets adultes jeunes et 25 sujets adultes âgés. Les réponses montrent une grande variabilité intra et intersujet. Les ondes 1, 3 et 5 sont des marqueurs constants et reproductibles de la réponse alors que les ondes 2, 4, 6 et 7 sont variables et fréquemment asymétriques ou absentes. Les latences de pics augmentent avec l'augmentation de la vitesse de stimulation, la diminution de l'intensité du stimulus et l'accroissement de l'âge. Les temps de conduction inter-pics, mis à part ceux qui impliquent l'onde 2, ne sont pas affectés par le changement d'intensité du stimulus. Le temps entre les ondes 1 et 3 augmente avec l'augmentation de vitesse de stimulation et l'accroissement en âge, alors que le temps entre les ondes 3 à 5 n'est affecté par aucun changement des paramètres du stimulus ni par l'âge. L'amplitude de l'onde seule n'est pas une mesure fiable de normalité.

Les valeurs normales pour les latences de pics, les temps de conduction inter-pics et la symétrie entre les deux oreilles sont présenté. Des intervalles de confiance à 95 et 99% sont obtenus pour les temps de conduction normaux inter-pics avec les corrections appropriées pour les vitesses de stimulation utilisées et l'âge des patients. Aucune correction n'est nécessaire pour les modifications d'intensité du stimulus. Ceci suggère qu'aucune correction n'est nécessaire lorsqu'on étudie des malades présentant des déficits auditifs de type conduction. Les laboratoires utilisant la même technique de stimulation paraissent obtenir des valeurs de temps de conduction normatif équivalents.

The author wishes to thank Chris Carlson, REEGT, for her assistance in preparing this study.

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